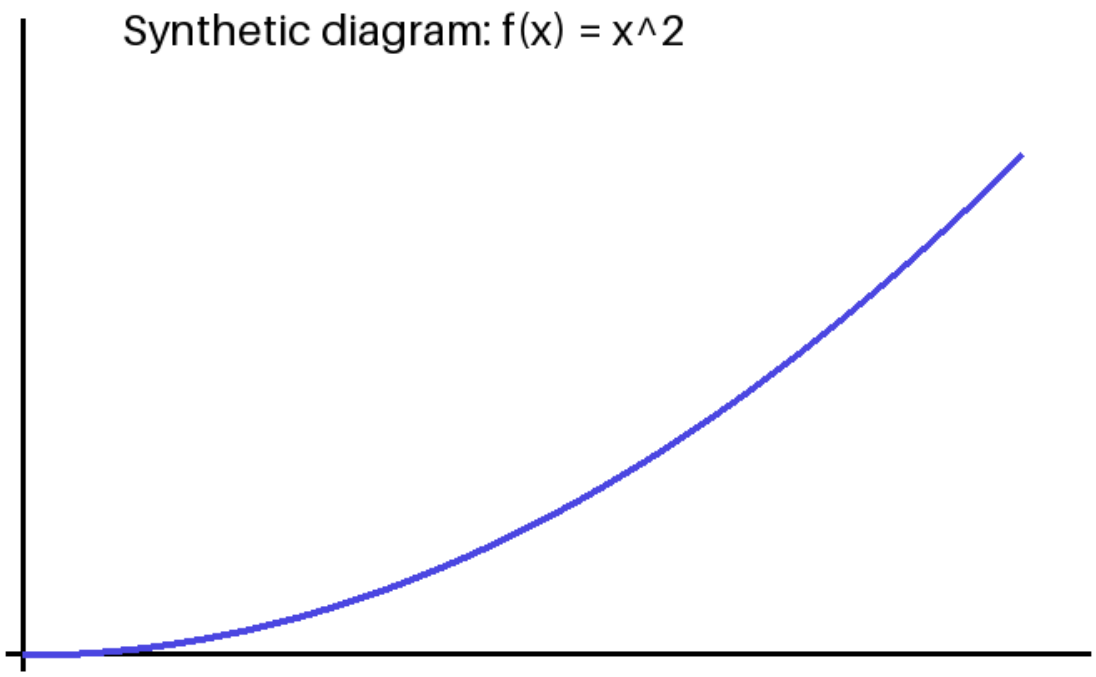
# Continuity on a compact interval

Definition. A real function f is continuous at a when, for every epsilon > 0, there exists delta > 0 such that |x-a| < delta implies |f(x)-f(a)| < epsilon.

Theorem. If f is continuous on the closed bounded interval [a,b], then f attains both a maximum and a minimum. Both endpoints are included.




Exercise: Why does f(x)=x on (0,1) have no maximum?